

Materials and Methods

Materials

Escherichia coli BL21(DE3) (provided by the instructor's lab) was used as the expression host, and the pET28 vector containing a kanamycin resistance gene was used as the selection marker.

For LB liquid medium, 10 g tryptone, 10 g NaCl, and 5 g yeast extract were dissolved in 1000 mL ddH₂O, and the medium was sterilized by high-pressure steam at 121 °C for 30 min. For LB solid medium, 10 g tryptone, 10 g NaCl, 5 g yeast extract, and 15 g agar were dissolved in 1000 mL ddH₂O, and the medium was sterilized by high-pressure steam at 121 °C for 30 min. Tryptone was purchased from Hangzhou Microbial Reagent Co., Ltd. (Hangwei), yeast extract was purchased from Oxoid (Oxoid™ Yeast Extract Powder), and agar was purchased from Qingdao Hope Bio-Technology Co., Ltd. (Hopebio). Kanamycin was supplemented at a final concentration of 50 µg/mL for selection when required.

Template DNA Extraction

A single colony was inoculated into 5 mL of LB liquid medium containing 50 µg/mL kanamycin and cultivated at 37 °C with shaking at 190 rpm until the OD₆₀₀ reached approximately 0.6. A 200 µL aliquot of the culture was transferred into a 1.5 mL EP tube and centrifuged at 12,000 rpm for 10 min to discard the supernatant. The cell pellet was resuspended in 100 µL of 1×TE buffer and thoroughly mixed until a uniform turbid suspension was obtained.

The resuspension was incubated in a boiling water bath for 10 min, followed by freezing at -20 °C for 30 min and immediate incubation on ice for 30 min. After the freeze-thaw treatment, the sample was centrifuged at 4 °C, 12,000 ×g for 10 min. The supernatant was collected as the template DNA. If not used immediately, the template

DNA was stored at -20 °C; prior to use, the sample was centrifuged again and the supernatant was collected for subsequent PCR amplification.

PCR Amplification

PCR amplification was performed using the extracted mutant plasmids as templates. Each 50 µL reaction mixture contained 10 µL of 5× GXL buffer, 4 µL of dNTP Mix, 0.5 µL of forward primer, 0.5 µL of reverse primer, 1 µL of GXL DNA polymerase, 1 µL of plasmid template, and 33.5 µL of ddH₂O.

The PCR program consisted of initial denaturation at 94 °C for 5 min, followed by 30 cycles of denaturation at 98 °C for 10 s, annealing at the primer-specific T_m for 15 s, and extension at 68 °C for 1 min per kb. After 30 cycles, a final extension was carried out at 68 °C for 10 min.

The PCR reaction mixture is shown below.

Reagent	Volume
5×GXL buffer	10 µL
dNTP Mix	4 µL
Forward primer	0.5 µL
Reverse primer	0.5 µL
GXL DNA polymerase	1 µL
Template DNA	1 µL
ddH ₂ O	33.5 µL
Total volume	50 µL

Table 2. PCR reaction mixture

Agarose Gel Electrophoresis

PCR products were separated by electrophoresis on 1% agarose gels. For the small, medium and large gels, 0.17 g, 0.34 g and 0.68 g agarose were dissolved in TAE buffer to final volumes of 17 mL, 34 mL and 68 mL, respectively. Gel stain was added at volumes of 1.7 μ L, 3.4 μ L and 6.8 μ L to the small, medium and large gels, respectively. The gel solutions were thoroughly mixed, poured into casting trays and allowed to solidify before electrophoresis.

Before loading, PCR samples were mixed with 6 $\times$ loading buffer. For 5 μ L of PCR product, 1 μ L of 6 $\times$ loading buffer was added. For a 50 μ L sample, 10 μ L of 6 $\times$ loading buffer was added. For standard wells, 3–4 μ L of sample was loaded per well, with a maximum loading volume of 5 μ L. Approximately 50 μ L of sample was loaded into large wells. DNA marker was loaded at 3 μ L per standard well or 6–7 μ L per large well. Electrophoresis was performed at 150–155 V for 20–25 min. After electrophoresis, the PCR products were visualized to confirm the presence of the target bands, and then subjected to sequencing verification to confirm the correct introduction of the mutation sites.

Induction and Sonication Lysis

The recombinant *E. coli* containing the expression vector was inoculated into LB liquid medium containing kanamycin and incubated at 37 °C with shaking until the OD₆₀₀ reached an appropriate value to obtain the seed culture. A 100 μ L aliquot of the seed culture was transferred into fresh LB liquid medium and incubated at 37 °C for 2.5 h, followed by incubation at 4 °C for 0.5 h. Protein expression was induced by adding 1 μ L of IPTG (final concentration 0.1 mM) at 16 °C for 18 h. Induced cells were harvested by centrifugation at 4 °C, 8000 rpm for 10 min, washed with fresh LB

medium, and resuspended in 1 mL of PBS buffer. The resuspension was lysed by sonication on ice. The sonication program was set as 5 s of sonication with a 10 s interval, lasting for 1 min in total, with a maximum of 3 cycles. The product was centrifuged and the supernatant was collected as the crude cell lysates.

HS-GC Analysis

The enzymatic reaction was carried out in a 200 μ L system containing crude cell lysates, SAM (1 mM), and methanethiol (MeSH, 1 mM)¹². The reaction mixtures were incubated at 28 °C for 3 h, followed by centrifugation to collect the supernatant. The supernatant was transferred into a headspace vial and sealed for HS-GC analysis.

GC analysis was performed on an Agilent 8860B gas chromatograph equipped with a headspace autosampler. Nitrogen was used as the carrier gas. The specific detector type, headspace oven temperature, injection port temperature, detector temperature, and column temperature program were set according to the standard operating procedure.

The peak areas of DMS and MeSH were integrated. The absolute product yields were quantified using the standard curve ($\sqrt{\text{peak area}} = 3.1854C - 25.261$) and expressed in nmol.

Data Collection and Preprocessing

The mutation data in this study were obtained from HS-GC experiments and stored in the "Wet-lab HS-GC data.xlsx" file, which contains the mutant names, the peak areas of DMS and MeSH, and the product yields (nmol). The peak areas were converted into absolute yields using the standard curve ($\sqrt{\text{peak area}} = 3.1854C - 25.261$). The yield of each variant was compared with the wild-type UE-2 to calculate the relative activity (%).

Computational Design

The three-dimensional structure of wild-type UE2 (MddH) was predicted using AlphaFold3. The first-round design (Route A) employed Boltz-2 to rank point mutants by binding affinity, identifying E63Q, E91Q, and T108I. The second-round design (Route B) was upgraded to a multidimensional consensus strategy. All residues within an 8 Å radius of the SAM-binding pocket were enumerated, yielding 855 single-point candidates. These candidates were evaluated by integrating three zero-shot fitness predictors (ProtSSN, ProSST, and TranceptEVE) with a ThermoMPNN stability red line ($\Delta\Delta G > +1$ kcal/mol rejected) and a PLACER ligand conformational gate. Based on the weighted consensus scores, the top-ranked variants were prioritized for future wet-lab validation, including I46L, E91Q, S65T, S44T, V40I, as well as the E91A/D/N variants designed for mechanistic probing.

The final consensus scores and candidate list were exported as a spreadsheet to guide subsequent experimental design. Based on the Route B consensus rankings, a prioritized shortlist of variants (e.g., I46L, E91Q, S65T, S44T, V40I, and E91A/D/N) was generated for future wet-lab validation. The complete ranking tables and raw computational outputs are provided in the Supplementary Material."

Data analysis and availability

HS-GC data processing

HS-GC data processing, statistical analysis, and plotting were performed using GraphPad Prism. The peak areas of DMS and MeSH obtained from HS-GC were converted into absolute product yields (nmol) using the standard curve ($\sqrt{\text{peak area}} = 3.1854C - 25.261$). Each variant contained at least three biological replicates, and the mean, standard deviation (SD), and coefficient of variation (CV) were calculated. Statistical significance was assessed using one-way ANOVA followed by Dunnett's

multiple comparison test against the wild-type UE-2. Data organization and plotting were performed using GraphPad Prism. The original data are provided as Supplementary Material.

Dry experiment project

The project code and computational data are currently available for download via the Wiki (Computational code and candidate rankings), and will be subsequently uploaded to GitHub and made publicly available.